

Efficient Algorithms for $\mathbb{G}_2$ Subgroup Membership Testing on Pairing-friendly Curves

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Abstract. Subgroup membership testing serves as a crucial countermeasure against small subgroup attacks, thereby ensuring the security of pairing-based cryptographic protocols. Despite its vital importance, the expensive computational requirements for membership testing on specific pairing-friendly curves pose a non-negligible challenge. In this paper, we revisit the $\mathbb{G}_2$ membership testing algorithms on KSS16 curves and propose a novel approach specifically designed for the families constructed by the KSS method (Kachisa-Schaefer-Scott method). Moreover, we generalize several previous methods for $\mathbb{G}_2$ membership testing, rendering them applicable to more generic pairing-friendly curves. Specifically, we implement an efficient $\mathbb{G}_2$ membership testing on three well-known curves KSS16-329, KSS16-330, and KSS16-766 for verification. The experimental results illustrate that our new method achieves improvements of 24.0%, 33.3%, and 29.2% in terms of clock cycles compared to the state-of-the-art, respectively.

Keywords: small subgroup attacks membership testing KSS curves endomorphisms

1 Introduction

In the realm of public-key cryptography, bilinear pairings on elliptic curves have become a significant component of many cryptographic protocols, such as identity-based encryption [10], authentication scheme [43], short signatures [11], zero-knowledge proof [18, 19], and blockchain-based systems [38]. The subgroups $\mathbb{G}_1$, $\mathbb{G}_2$ and $\mathbb{G}_T$ are three cyclic groups with large prime order r involved in a

pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, which is a bilinear map on an elliptic curve E . Denote by k and $\mathbb{F}_{p^k}$ the embedding degree of E with respect to r and a finite field of p^k elements, respectively. Within this framework, the two input groups $\mathbb{G}_1$ and $\mathbb{G}_2$ emerge as the subgroups of $E(\mathbb{F}_{p^k})$, while the output group $\mathbb{G}_T \subseteq \mathbb{F}_{p^k}$. The security of pairing-based protocols relies on the difficulty of solving discrete logarithm problems (DLP) in these three subgroups mentioned above [17,27,41].

Nevertheless, if the communication channel is insecure, the attackers may force the running system to return a point with a small order. Thus, there is a potential probability that the cryptographic systems suffer from the small-subgroup attacks [6,37]. Two feasible countermeasures, named cofactor clearing and subgroup membership testing, have been proposed to avert such vulnerabilities.

The cofactor clearing is to multiply the target points Q hashed on the corresponding curves [21] by the cofactor to make the candidate elements within the expected subgroups of order r . This approach is efficient when working on $\mathbb{G}_1$ since the corresponding cofactor h_1 is relatively small. In 2022, Housni, Guillevic, and Piellard investigated cofactor clearing and proposed an efficient algorithm on $\mathbb{G}_1$ [20]. In the situation of $\mathbb{G}_2$, the cofactor h_2 becomes crucially large. Even though the tools in [13,23,34,40] can be exploited to reduce the sizes of scalar multiplications, this algorithm is still cumbersome. Furthermore, cofactor clearing may bring several extra problems since it necessitates alterations to the parameters of the protocols, often impeding the implementation [39].

Troubles arising from cofactor clearing can be circumvented by performing subgroup membership testing, which is designed to determine whether the candidate element has prime order r . A naive approach is to multiply the element by r (raise the element to the power of r in the case of $\mathbb{G}_T$) and verify its equivalence to the identity element $\mathcal{O}_E \in E(\mathbb{F}_{p^k})$ ($1 \in \mathbb{F}_{p^k}$ in the case of $\mathbb{G}_T$). However, this method leads to a huge computational cost and significantly weakens the performance of the whole protocol. In 2021, Scott [39] introduced efficient algorithms for the subgroup membership testing for $\mathbb{G}_1$, $\mathbb{G}_2$ and $\mathbb{G}_T$ on the BLS family [7]. This approach was subsequently generalized by Housni, Guillevic and Piellard [20] to the BN family [8]. In 2023, Dai *et al.* [16] proposed new techniques of the membership testing (DLZZ criterion [3]) for $\mathbb{G}_1$, $\mathbb{G}_2$ and $\mathbb{G}_T$, which are suitable for a broader class of pairing-friendly curves with j -invariants 0 or 1728, including BN, BLS and BW [22] families. Koshelev [35] leveraged the information of the cofactor h and proposed efficient techniques for testing the memberships of points in the prime-order subgroups of elliptic curves over finite fields by employing Tate pairings, under the condition that h is relatively small. More recently, based on Koshelev's method, Dai *et al.* [14] proposed a new approach to accelerate the $\mathbb{G}_1$ membership testing on BLS and BW curves.

In recent years, the demand for high-assurance pairing-based cryptographic systems has extended beyond the 128-bit security level offered by popular families of pairing-friendly curves such as BN and BLS12. As applications including identity-based encryption [10] and advanced zero-knowledge proof frameworks [18,19] evolve toward stronger security guarantees and more complex multi-

pairing computations, elliptic curves with higher embedding degrees have become increasingly relevant. Within this context, the Kachisa-Schaefer-Scott (KSS) [31] family, particularly family KSS16 offers a well-balanced trade-off between high security (approximately 192-bit) and practical efficiency in extension-field arithmetic. Aranha *et al.* [4] demonstrated that optimized pairing implementations at the 192-bit security level are practical, identifying implementation techniques and candidate curves, including KSS16, suitable for high-security deployment. Moreover, in light of advances in discrete logarithm algorithms, particularly the extended Number Field Sieve (exTNFS) [33], Barbulescu and Duquesne’s [5] revised security estimates under the exTNFS attack framework and provided updated parameter recommendations for several major families, including KSS16. Their results show that KSS16 remains a viable candidate at the 192- and even 256-bit security levels when appropriate parameter sizes are selected, thereby reinforcing its relevance for applications that require long-term or high-assurance security guarantees.

The KSS16 curves have been identified as a promising candidate for cryptographic protocols that require multiple or chained bilinear operations, such as proof aggregation and recursive verification in modern proof systems [46]. Recent literature underscores this relevance. Zhang and Lin [46] concluded that KSS16 is among the most efficient choices for computing products/quotients of pairings at the 192-bit level. Aranha, Housni, and Guillevic [2] highlighted the emerging importance of family KSS16 for succinct non-interactive arguments of knowledge (SNARKs) and other proof frameworks. They emphasize that, beyond raw pairing efficiency, such pairing-friendly curves must support composability, high 2-adicity, and favorable extension-field structures, which demonstrates strong potential. Besides, KSS16 curves have been incorporated in several well-known high-performance cryptographic libraries, such as RELIC [1] and ELiPS [32] for the research at high security levels. In conclusion, KSS16 represents an effective alternative ext-generation cryptographic protocols that integrate proof composition, privacy-preserving computation, and long-term cryptographic soundness within a unified, pairing-friendly framework. Up to now, there has been a lack of research that systematically focuses on the $\mathbb{G}_2$ membership testing on the KSS family. Motivated by this gap, we explore deriving an efficient approach for the $\mathbb{G}_2$ membership testing tailored to the KSS family.

Our Contributions. In this paper, we revisit the previous methods for $\mathbb{G}_2$ membership testing, then propose some new and more extensive algorithms. We first provide an efficient algorithm for the $\mathbb{G}_2$ membership testing on the KSS family [31] admitting fast pairing computations. Moreover, we extend Scott’s method and DLZZ criterion to more generic families of pairing-friendly curves. In more detail, our contributions are summarized as follows:

1. We first concentrate on the KSS family and develop an effective algorithm for $\mathbb{G}_2$ membership testing that leverages some specific properties of this family. Our algorithm is tailored to family KSS16, which is one of the efficient choices for pairing computations on curves at 128-bit security level [5]. In particular,

our new method demonstrates a substantial improvement in performance over DLZZ criterion on two popular pairing-friendly curves KSS16-330 [1] and KSS16-329 [2], that can be suitable for SNARKs [18,19] (Succinct Non-interactive ARguments of Knowledge).

2. We generalize Scott’s method for the $\mathbb{G}_2$ membership testing by relaxing the corresponding condition, making it applicable to a wider class of pairing-friendly curves. This generalization outperforms DLZZ criterion on curve BLS27-426 ensuring the 192-bit security level [3], which admits an efficient $\mathbb{G}_1$ arithmetic among 192-bit security curves and possesses potential to be utilized in DAA (Direct Anonymous Attestation) schemes [44].
3. Moreover, we extend the techniques proposed by Dai *et al.* to any families of GLV-curves [26] equipped with degree- n endomorphisms ($n \geq 2$) besides the curves with CM-discriminants 1 or 3, and give the condition that determines whether the computational cost can be reduced to $\log_2(r)/2\varphi(k)$ bit operations. This idea can also be employed to extend DLZZ criterion for $\mathbb{G}_1$ membership testing on more pairing-friendly curves.
4. Finally, we implement our methods on three KSS16 curves in RELIC. The experimental results illustrate that our new methods obtain an enhancement of about **24.0%**, **33.3%** and **29.2%** in terms of CPU clock cycles compared to DLZZ criterion for the $\mathbb{G}_2$ membership testing on KSS16-329, KSS16-330 and KSS16-766, respectively. Our source code is available at

<https://github.com/SMtestings/relic-G2MT-KSS16>.

Organizations of this paper. The remainder of this paper is outlined as follows. Section 2 presents the definitions of the pairing subgroups, GLV-endomorphisms of ordinary elliptic curves, along with the KSS construction. The efficient methods for the $\mathbb{G}_2$ membership testing are described in Sections 3. In Section 4, we provide the concrete cost analysis and experimental results. The conclusion is presented in Section 5.

2 Mathematical Background

In this section, we first introduce the fundamental concepts of the three pairing subgroups $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$. Subsequently, we provide a concise overview of the efficiently computable endomorphisms on ordinary curves. Moreover, we also briefly present the KSS method [31].

2.1 Pairing subgroups on ordinary elliptic curves

Denote by E an ordinary elliptic curve over a finite field $\mathbb{F}_p$, where p and $\mathcal{O}_E$ are a large prime and the identity element in E , respectively. Let $\#E(\mathbb{F}_p)$ represent the group order of $E(\mathbb{F}_p)$. Then we have $\#E(\mathbb{F}_p) = p + 1 - t$ [42, Chapter 4.3], with t the trace of p -power Frobenius endomorphism $\pi : (x, y) \rightarrow (x^p, y^p)$ satisfying the characteristic equation

$$\pi^2 - t\pi + p = 0.$$

Let r be a large prime factor of $\#E(\mathbb{F}_p)$. Define the embedding degree k with respect to r to be the smallest positive integer satisfying $r \mid p^k - 1$. We denote the eigenspaces of π acting on the r -torsion group $E[r] = \{P \in E \mid [r]P = \mathcal{O}_E\}$ with eigenvalues 1 and p as $\mathbb{G}_1$ and $\mathbb{G}_2$, respectively. More precisely, these two pairing subgroups can be expressed as:

$$\mathbb{G}_1 = E(\mathbb{F}_p)[r], \quad \mathbb{G}_2 = E(\mathbb{F}_{p^k})[r] \cap \ker(\pi - [p]).$$

Furthermore, if E admits a twist E' over $\mathbb{F}_{p^e}$ with $e = k/d$, where d denotes the order of the automorphism group $\text{Aut}(E)$. Define $\phi : E' \rightarrow E$ to be the twisting isomorphism. Then $\mathbb{G}_2$ can be concisely represented as $\mathbb{G}_2 = E'(\mathbb{F}_{p^e})[r]$. On this basis, the straightforward methods for $\mathbb{G}_2$ membership testing are presented as follows

1. For curves admitting a twist: $Q \in \mathbb{G}_2 = E'(\mathbb{F}_{p^e})[r] \Leftrightarrow Q \in E'(\mathbb{F}_{p^e})$ and $[r]Q = \mathcal{O}_{E'}$.
2. For curves without a twist: $Q \in \mathbb{G}_2 = E(\mathbb{F}_{p^k})[r] \cap \ker(\pi - [p]) \Leftrightarrow Q \in E(\mathbb{F}_{p^k}), [r]Q = \mathcal{O}_E$ and $\Phi_k(\pi)(Q) = \mathcal{O}_E$,

where Φ_k denotes the k -th cyclotomic polynomial.

2.2 Efficiently computable endomorphisms of ordinary curves

In this subsection, we introduce the efficiently computable endomorphisms of ordinary curves. Define D as a positive square-free integer such that $4p - t^2 = Dy^2$, $y \in \mathbb{Z}$. According to [42, Chapter 10.2], the endomorphism ring of an ordinary curve E over a finite field is isomorphic to an order in an imaginary quadratic field $\mathbb{Q}(\sqrt{-D})$. The largest subring of $\mathbb{Q}(\sqrt{-D})$ is

$$\mathcal{O}_K = \begin{cases} \mathbb{Z} \left[\frac{1+\sqrt{-D}}{2} \right] & \text{if } D \equiv 3 \pmod{4}, \\ \mathbb{Z} [\sqrt{-D}] & \text{if } D \equiv 1, 2 \pmod{4}. \end{cases}$$

An **order** $R = \mathbb{Z} + \mathbb{Z}f\delta$ in $\mathbb{Q}[\sqrt{-D}]$ is a subring of $\mathcal{O}_K$ such that $\mathbb{Z} \subsetneq R \subseteq \mathcal{O}_K$ [42, Chapter 10.2], where $f > 0$ and $\delta = \frac{1+\sqrt{-D}}{2}$ or $\sqrt{-D}$. Let $\sigma \in \text{End}_p(E)$ be an endomorphism of E over $\mathbb{F}_p$. Thus σ can be represented as $\sigma = a + b\sqrt{-D}$, where $a, b \in \mathbb{Q}$ and $2a, 2b \in \mathbb{Z}$. Define $\text{Nrd}(\sigma) = \sigma\bar{\sigma} = (a + b\sqrt{-D})(a - b\sqrt{-D})$ to be the norm of σ .

An endomorphism σ with norm n is named as a degree- n endomorphism, which satisfies the following characteristic equation

$$\sigma^2 - 2a\sigma + n = 0. \tag{1}$$

This morphism enables scalar multiplication by λ_1 (resp. λ_2) on points in $\mathbb{G}_1$ (resp. $\mathbb{G}_2$), where λ_1 and λ_2 are the two distinct roots of the above equation. A degree-1 endomorphism of E is exactly the automorphism.

In the following, we introduce some common endomorphisms of ordinary curves. First, consider an ordinary curve $E : y^2 = x^3 + b$, $b \in \mathbb{F}_p$ with $D = 3$

and $p \equiv 1 \pmod{3}$. There is an endomorphism $\sigma : (x, y) \rightarrow (\alpha x, y)$ on E with $\sigma = \frac{1+\sqrt{-3}}{2}$, where α is a primitive cube root of unity in $\mathbb{F}_p^*$. According to Eq. (1), σ satisfies $\sigma^2 + \sigma + 1 = 0$.

Similarly, for an elliptic curve $E : y^2 = x^3 + ax$, $a \in \mathbb{F}_p$ with $D = 1$ and $p \equiv 1 \pmod{4}$, there exists an endomorphism $\sigma : (x, y) \rightarrow (-x, \beta y)$ on E , where β is a fourth primitive root of unity in $\mathbb{F}_p^*$. This efficiently computable endomorphism satisfies the characteristic equation $\sigma^2 + 1 = 0$ since $\sigma = \sqrt{-1}$.

These two endomorphisms allow for efficient scalar multiplications by employing the GLV method [26]. We refer to such endomorphisms as GLV-endomorphisms. A curve equipped with such an endomorphism is called a GLV-curve.

Additionally, the twisting isomorphism ϕ induces an endomorphism $\psi = \phi^{-1} \circ \pi \circ \phi$ [25] on E' , satisfying the corresponding characteristic equation $\psi^2 - t\psi + p = 0$. The action of ψ on $Q \in \mathbb{G}_2 = E'(\mathbb{F}_{p^e})[r]$ allows for a scalar multiplication $[t-1]$. This endomorphism has been used for enhancing the efficiency of scalar multiplications in $\mathbb{G}_2$ [25].

2.3 The KSS construction

Kachisa, Schaefer and Scott [31] proposed a method (KSS method) for constructing pairing-friendly curves based on the Brezing-Weng method (BW method) [12]. Using the BW method with a non-cyclotomic polynomial, the authors in [31] constructed a list of pairing-friendly curves with embedding degrees 16 and 18. In the following, we briefly describe this method.

Let $r(x)$, $t(x)$ and $p(x)$ respectively denote the parameterized polynomials of the prime factor r , the Frobenius trace t and the characteristic p . We set the number field $K = \mathbb{Q}(\zeta_k, \sqrt{-D})$, which contains the primitive k -th root ζ_k and $\sqrt{-D}$.

The polynomial $r(x)$ is defined to be the minimal polynomial of $\alpha = (a + b\sqrt{-D}) \cdot \zeta_k \in K$. Let $t(x)$, $y(x) \in \mathbb{Q}[x]/(r(x))$ be polynomials corresponding to $\zeta_k + 1$ and $\frac{\zeta_k - 1}{\sqrt{-D}}$ in K , respectively. Then $p(x) = \frac{t^2(x) + Dy^2(x)}{4}$ represents the parameterized polynomial of p such that

$$r(x) \mid p(x) + 1 - t(x) \text{ and } r(x) \parallel p^k(x) - 1.$$

Therefore, these three polynomials define a family of pairing-friendly curves. We present a well-known family called KSS16 for illustration.

Example 1 (KSS16). A family constructed from $\alpha = (2\sqrt{-1} - 1) \cdot \zeta_{16}$ with $D = 1$ and $k = 16$.

$$t(x) = \frac{1}{35}(2x^5 + 41x + 35),$$

$$r(x) = x^8 + 48x^4 + 625,$$

$$p(x) = \frac{1}{980}(x^{10} + 2x^9 + 5x^8 + 48x^6 + 152x^5 + 240x^4 + 625x^2 + 2398x + 3125).$$

According to [5], KSS16 is one of the popular choices for the optimal ate pairing computation on curves achieving the 128-bit security level.

3 Main Results

In this section, we present our methods of subgroup membership testing for $\mathbb{G}_2$ on pairing-friendly curves. We first provide a fast testing method on KSS curves. Besides, we extend the methods proposed in [39] and [16] to more pairing-friendly curves. In the remainder of this paper, we only consider the curve E admitting a degree- d twist E' defined over $\mathbb{F}_{p^{k/d}} = \mathbb{F}_{p^e}$. Denote by h_1 and h_2 the cofactors $\#E(\mathbb{F}_p)/r$ and $\#E'(\mathbb{F}_{p^e})/r$, respectively.

3.1 An efficient $\mathbb{G}_2$ membership testing on KSS curves

In this subsection, we focus on the pairing-friendly curves constructed by the KSS method and give an efficiently tailored approach for the $\mathbb{G}_2$ -membership testing on such curves. Moreover, we present the corresponding computation analysis and comparison on a specific family of curves called KSS16.

Recalled from Section 2.2, a degree- n endomorphism $\sigma \in \text{End}_p(E)$ can be represented as $\sigma = a + b\sqrt{-D}$, where $a, b \in \mathbb{Q}$ and $a^2 + b^2D = n$. The KSS family is constructed by making $r(x)$ the minimal polynomial of the element

$$\alpha = (a + b\sqrt{-D}) \cdot \zeta_k.$$

Consequently, assume that E is a KSS curve admitting a degree- d twist, then for every $Q \in \mathbb{G}_2 = E'(\mathbb{F}_{p^e})[r]$ it yields that

$$\tau(Q) = [x]Q,$$

where the map $\tau = \phi^{-1} \circ \sigma \circ \pi \circ \phi$ and x are the composition of Φ , ψ , and the parametrized seed of E , respectively. Thus τ corresponds to the element $\alpha = (a + b\sqrt{-D}) \cdot \zeta_k \in \mathbb{Q}(\zeta_k, \sqrt{-D})$. Using the above notations, we present our approach for the $\mathbb{G}_2$ membership testing on KSS curves, which is summarized in the following theorem.

Theorem 1. *Let E be a KSS curve admitting a degree- d twist E' . The parameterized polynomial $r(x)$ of E is the minimal polynomial of the element $\alpha = (a + b\sqrt{-D}) \cdot \zeta_k$. Denote by f the integer satisfying the CM equation $Df^2 = 4p - t^2$. Let $\gcd((x^2 + (at + bfD)x + np)/r, h_2) = s$. Define $\{P_1, P_2\}$ to be a basis such that $\langle P_1, P_2 \rangle = E'[s]$.*

1. *If $s = 1$, then we have $Q \in \mathbb{G}_2 \Leftrightarrow Q \in E'(\mathbb{F}_{p^e})$ and $\tau(Q) = [x]Q$.*
2. *If $s \nmid h_2$, assume that $\langle P_1 \rangle = E'(\mathbb{F}_{p^e})[s]$. If the point $Q \in E'(\mathbb{F}_{p^e})$ satisfies that $t_s(P_1, Q) \neq 1$ where $t_s(\cdot, \cdot)$ represents the reduced Tate pairing, then $Q \notin \mathbb{G}_2$. Otherwise, we have*

$$Q \in \mathbb{G}_2 \Leftrightarrow Q \in E'(\mathbb{F}_{p^e}) \text{ and } \tau(Q) = [x]Q.$$

3. *If $s^2 \mid h_2$, and the point $Q \in E'(\mathbb{F}_{p^e})$ satisfies that $t_s(P_1, Q) \neq 1$ or $t_s(P_2, Q) \neq 1$, then $Q \notin \mathbb{G}_2$. Otherwise, we have*

$$Q \in \mathbb{G}_2 \Leftrightarrow Q \in E'(\mathbb{F}_{p^e}) \text{ and } \tau(Q) = [x]Q.$$

To prove this theorem, we first consider the characteristic polynomial of τ and present the following lemma, which is the generalization of [15, Lemma 2] for description.

Lemma 1. *Let E denote a GLV-curve, and $\Phi = \phi^{-1} \circ \sigma \circ \phi$ an degree- n endomorphism associated with $a + b\sqrt{-D} \in \text{End}_p(E')$, $a, b \in \mathbb{Q}$. Then the characteristic equation of $\tau = \Phi \circ \psi$ is given by*

$$\tau^2 + (at \pm b f D)\tau + np, \quad (2)$$

where t and f represent the Frobenius trace and an integer such that $Df^2 = 4p - t^2$, respectively.

Proof. First, we assume that the characteristic equation is $\tau^2 + M\tau + N = 0$. It can be derived that

$$\begin{aligned} N &= \tau\bar{\tau} = \deg(\tau) = \deg(\Phi) \cdot \deg(\psi) = np, \\ M &= -(\tau + \bar{\tau}). \end{aligned}$$

According to the characteristic equation of $\psi = \phi^{-1} \circ \pi \circ \phi$, we obtain

$$\psi = \frac{t \pm f\sqrt{-D}}{2}.$$

Consequently, the composited map τ can be represented as

$$\tau = \Phi \circ \psi = (a + b\sqrt{-D}) \cdot \frac{t \pm f\sqrt{-D}}{2} = \frac{at \mp b f D \pm (af + bt)\sqrt{-D}}{2},$$

which implies that $M = at \pm b f D$. □

For simplicity, we assume that the characteristic equation of τ is $\tau^2 + (at + b f D)\tau + np$ in the remainder of this paper. Thus for every $Q \in E'(\mathbb{F}_{p^e})$, it yields that

$$\tau^2(Q) + [at + b f D]\tau(Q) + [np]Q = \mathcal{O}_E, \quad (3)$$

Based on the above analysis, we can deduce that

$$[x^2 + (at + b f D)x + np]Q = \mathcal{O}_E$$

for every $Q \in \mathbb{G}_2$, which implies that

$$x^2 + (at + b f D)x + np \equiv 0 \pmod{r}.$$

By [42, Theorem 4.1] we know that the group $E'(\mathbb{F}_{p^e})$ can be represented as

$$E'(\mathbb{F}_{p^e}) \cong \mathbb{Z}_{m_1} \oplus \mathbb{Z}_{m_1 n_1 r},$$

where $m_1, n_1 \in \mathbb{N}$. Thus, it yields that $h_2 = m_1^2 n_1$. Using these notations, we provide the proof process of Theorem 1.

Proof of Theorem 1. The necessity is obvious. We now prove the sufficiency.

If $s = 1$, then we can derive that

$$\gcd(x^2 + (at + bfD)x + np, h_2r) = r.$$

If it satisfies $[x^2 + (at + bfD)x + np]Q = \mathcal{O}_E$ for $Q \in E'(\mathbb{F}_{p^e})$, we deduce that

$$\text{ord}(Q) \mid x^2 + (at + bfD)x + np.$$

By $\gcd(x^2 + (at + bfD)x + np, h_2r) = r$ and $\#E'(\mathbb{F}_{p^e}) = h_2r$, we have

$$\text{ord}(Q) = r \Leftrightarrow Q \in \mathbb{G}_2.$$

Substituting $[np]Q = [-(at + bfD)x - x^2]Q$ into Eq. (3) yields

$$\tau^2(Q) + [at + bfD]\tau(Q) - [(at + bfD)x + x^2]Q = \mathcal{O}_{E'},$$

which is equivalent to

$$\tau(Q) = [x]Q \text{ or } \tau(Q) = [-at - bfD - x]Q.$$

If $\tau(Q) = [-at - bfD - x]Q$, it follows from

$$\gcd(x^2 + (at + bfD)x + np, h_2r) = r$$

that $Q \in \mathbb{G}_2$, which is contradictory to $\tau(Q) = [x]Q$. Consequently, we deduce that

$$\tau(Q) = [x]Q \Leftrightarrow Q \in \mathbb{G}_2.$$

When $s \neq 1$, there are the following two cases.

1. If $s \mid h_2$, then it can be derived that $s \mid n_1$ and $E'(\mathbb{F}_{p^e})[s] \cong \mathbb{Z}_s$. Thus there exists a point $P_1 \in E'(\mathbb{F}_{p^e})$ such that $E'(\mathbb{F}_{p^e})[s] = \langle P_1 \rangle$. Every point $P \in E'(\mathbb{F}_{p^e})[s]$ can be represented as $P = [c]P_1$, $c \in \mathbb{Z}$. If the reduced Tate pairing $t_s(P_1, Q) \neq 1$, by [24, Theorem 26.3.3] we have $Q \notin sE'(\mathbb{F}_{p^e})$, which implies that $Q \notin \mathbb{G}_2$. If $t_s(P_1, Q) = 1$, then it yields that

$$t_s(P, Q) = t_s([c]P_1, Q) = t_s(P_1, Q)^c = 1$$

for any point $P \in E'(\mathbb{F}_{p^e})[s]$. By [24, Theorem 26.3.3] it holds that $Q \in sE'(\mathbb{F}_{p^e})$. Consequently, the cofactor h_2 can be shortened to

$$h'_2 = \#sE'(\mathbb{F}_{p^e})/r = h_2/s,$$

and we deduce that $\gcd(h_1, h'_2) = \gcd(h_1, h_2)/s = 1$. This means that

$$\tau(Q) = [x]Q \Leftrightarrow Q \in \mathbb{G}_2.$$

2. If $s^2|h_2$, we have $s|m_1$ and $E'[s] \subseteq E'(\mathbb{F}_{p^e})$. Hence, there exists a basis $\{P_1, P_2\}$ such that $E'(\mathbb{F}_{p^e}) = \langle P_1, P_2 \rangle$. If $t_s(P_1, Q) \neq 1$ or $t_s(P_2, Q) \neq 1$, by [24, Theorem 26.3.3] we deduce that $Q \notin sE'(\mathbb{F}_{p^e})$, thus $Q \notin \mathbb{G}_2$. Otherwise, for every $P \in E'(\mathbb{F}_{p^e})[s]$ we have

$$t_s(P, Q) = t_s([c_1]P_1 + [c_2]P_2, Q) = t_s(P_1, Q)^{c_1} \cdot t_s(P_2, Q)^{c_2} = 1,$$

which implies that $Q \in sE'(\mathbb{F}_{p^e})$. The remaining proof is similar to the situation $s||h_2$, and we complete the proof of Theorem 1. $\square$

Inspired by the above main theorem, we present Algorithm 1 for subgroup membership testing. It is remarked that in Algorithm 1 we only consider the first situation $s||h_2$ when $s > 1$, and the other situation is similar. If the tested point Q is in $\mathbb{G}_2$, then the algorithm returns true. Otherwise, it returns false.

Algorithm 1 $\mathbb{G}_2$ membership testing for the KSS family

Input: The point $Q \in E'(\mathbb{F}_{p^e})$ for the $\mathbb{G}_2$ membership testing. The cofactor h_2 . A precomputed s -torsion point P , where $s = \gcd((x^2 + (at + bfD)x + np)/r, h_2)$.

Output: Whether Q is in $\mathbb{G}_2$.

```

1: if  $s = 1$  then
2:   if  $\tau(Q) = [x]Q$  then
3:     return true
4:   else
5:     return false
6:   end if
7: else
8:    $T \leftarrow t_s(P, Q)$ 
9:   if  $T \neq 1$  then
10:    return false
11:  else
12:    if  $\tau(Q) = [x]Q$  then
13:      return true
14:    else
15:      return false
16:    end if
17:  end if
18: end if

```

Note that the generator(s) P_i of the subgroup $E'(\mathbb{F}_{p^e})[s]$ can be precomputed. The cost of the Algorithm 1 may involve computing the reduced Tate pairing(s) $t_s(P_i, Q)$, an endomorphism action τ , and a scalar multiplication by x . If s is small, the pairing computation can be roughly regarded as the computation of the s -th power residue symbol. As discussed in [30,35], the symbol can be determined by the reciprocity law with a complexity of $\mathcal{O}((\log_2 q)^2)$ for $s \leq 11$, where $q = p^e$. If so, the scalar multiplication $[x]$ is the main time-consuming part and our algorithm will exhibit greater efficiency.

We provide the following example (KSS16 family) for illustration, and make a concrete computational cost comparison between DLZZ criterion [16] and ours.

Example 2 (KSS16 family). Recalled from Example 1 in Section 2.3, the KSS16 family with CM-discriminant $D = 1$ is constructed by making $r(x)$ be the minimum polynomial of $\alpha = (2\sqrt{-1} - 1) \cdot \zeta_{16} \in \mathbb{Q}(\zeta_{16})$. Assume that E is a KSS16 curve, then it admits a degree-4 twist E' over $\mathbb{F}_{p^4}$. Consequently, it can be deduced that

$$([2]\sigma - [1]) \circ \pi(Q) = [x]Q,$$

for every $Q \in \mathbb{G}_2 = E'(\mathbb{F}_{p^4})$, where σ is an efficiently computable automorphism

$$\sigma : (x, y) \mapsto (-x, iy), \quad i^2 = -1.$$

The testing method in [16] is to utilize Frobenius endomorphism π and LLL's algorithm [36]. According to [16, Example 2] and [3, Section 4.3.2], there exists a short vector for $\mathbb{G}_2$ membership testing when $x \equiv 25, 45 \pmod{70}$. The vector $(c_0, \dots, c_7)$ such that $\sum_{i=0}^7 [c_i] \psi^i(Q) = \mathcal{O}_{E'}$ is illustrated as follows

$$(c_0, c_1, \dots, c_7) = (11u + 4, -9u - 3, 3u + 1, 3u + 1, -13u - 5, 7u + 3, u, 11u + 4),$$

where $u = \frac{-x-25}{70}$. Then by [16, Theorem 1], the testing of Q is to perform the following determination

$$Q \in \mathbb{G}_2 \Leftrightarrow \begin{cases} Q \in E'(\mathbb{F}_{p^4}), \\ \sum_{i=0}^6 \psi^i([c_i]Q) = -\psi([c_7]Q). \end{cases}$$

Since the cost of checking $Q \in E'(\mathbb{F}_{p^4})$ is relatively low, we neglect it for simplicity. The above testing requires a scalar multiplication by u , 13 point additions, one point doubling together with seven actions of the endomorphism ψ .

We now analyze the computational process of our testing method. By Lemma 1 we can see that the characteristic equation of $\tau = \phi^{-1} \circ ([2]\sigma - [1]) \circ \pi \circ \phi$ is

$$\tau^2 + (t - 2f)\tau + 5p = 0.$$

We can determine the polynomial $h_2(x)$ by [29, Proposition 2]. Using the condition $x \equiv 25, 45 \pmod{70}$ and the polynomial valuation theory, it yields that

$$2 \parallel h_2(x)r(x) = \#E'(\mathbb{F}_{p^4}).$$

Therefore, we have

$$\gcd(x^2 + (t - 2f)x + 5p, h_2(x)r(x)) = 2.$$

By Theorem 1, it requires to perform the reduced Tate pairing computation and determine whether

$$t_2(P_2, Q) = (x_Q - x_{P_2})^{\frac{p^4-1}{2}} = 1,$$

where P_2 is a precomputed 2-torsion point in $E'(\mathbb{F}_{p^4})$. By the equation of E' and $2 \parallel \#E'(\mathbb{F}_{p^4})$, we know that there is only one 2-torsion point $P_2 = (0, 0)$ in $E'(\mathbb{F}_{p^4})$. Consequently, it is equivalent to compute the quadratic residue symbol of the x -coordinate of Q and check whether

$$\left(\frac{x_Q}{p^4}\right)_2 = 1,$$

which can be accomplished by using the law of quadratic reciprocity. Thus, the corresponding computation cost of this part is negligible. If we have $\left(\frac{x_Q}{p^4}\right)_2 = 1$, then by Theorem 1 it remains to check

$$Q \in \mathbb{G}_2 \Leftrightarrow \begin{cases} Q \in E'(\mathbb{F}_{p^4}), \\ \tau(Q) = ([2]\sigma - [1]) \circ \phi^{-1} \circ \pi \circ \phi(Q) = [x]Q, \end{cases}$$

which takes about a scalar multiplication by x , a point addition, a point doubling, an action of ψ , an action of σ and a quadratic residue symbol to execute. Otherwise, we claim that $Q \notin \mathbb{G}_2$. The detailed comparison of the operations required for DLZZ criterion [16] and our method on the KSS16 family is illustrated in Table 1. From Table 1, we can see that our method saves about 12 point additions in $E'(\mathbb{F}_{p^4})$ compared to DLZZ criterion, since the costs of scalar multiplications are close and the other operations are relatively inexpensive. In conclusion, our new algorithm outperforms DLZZ criterion on any KSS16 curve with respect to the seed $x \equiv \pm 25 \pmod{70}$, including two well-known pairing-friendly curves: SNARKs-curve [18, 19] KSS16-329 [2], and curve KSS16-330 integrated into the cryptographic library RELIC [1].

Now we make a more concrete cost analysis on curve KSS16-330 associated with the seed $x = 0x3c4803fe1$ [5] at the 128-bit security level. Since the actions of endomorphisms and the computation of the residue symbol are very fast, we mainly consider the cost of the other operations. By utilizing the technique of NAF expansion, the scalar multiplication $[u]$ needs 29 point doublings and 11 point additions, while $[x]$ requires to execute 35 point doublings and 7 point additions. Thus, the total costs of DLZZ criterion and Theorem 1 are about 30 point doublings with 24 point additions, and 36 point doublings with 8 point additions, respectively. According to [9], the computational cost of a point addition (resp. a point doubling) on $E'(\mathbb{F}_{p^4})$ utilizing projective coordinates is 12 multiplications (resp. 8 multiplications together with 3 squarings) over $\mathbb{F}_{p^4}$, respectively. Hence, it requires to execute 528 multiplications with 90 squarings, and 384 multiplications with 108 squarings over $\mathbb{F}_{p^4}$ by employing DLZZ criterion and our approach, respectively. This verifies that our method needs fewer operations to perform than that of DLZZ criterion.

3.2 Generalizing Scott's method for $\mathbb{G}_2$ membership testing on more pairing-friendly curves

In this subsection, we aim to derive $\mathbb{G}_2$ membership testing methods that are applicable for more pairing-friendly curves. For the approach proposed by Scott

Table 1. The operations involved in our method and the approach in [16] for the $\mathbb{G}_2$ membership testing on the KSS16 curves. The notations PDBL and PADD denote the point doublings and additions using projective coordinates, respectively.

Family	This work	[16]
KSS16	1 scalar $[x]$	1 scalar $[\frac{-x-25}{70}]$
	1 PDBL + 1 PADD	1 PDBL + 13 PADD
	1 action of ψ + 1 action of σ	7 actions of ψ
	1 quadratic residue symbol $\left(\frac{x_Q}{p^4}\right)_2$	

[39], we weaken the corresponding condition and provide a generalized version which is different from the method in [16].

An efficient method for $\mathbb{G}_2$ membership testing on BN and BLS curves was first proposed by Scott in [39]. This testing method needs to satisfy the condition $\gcd(h_1, h_2) = 1$. The most expensive part of the computation of Scott's approach is the scalar multiplication by $t - 1$, where t is the Frobenius trace. However, in several situations $\gcd(h_1, h_2) = 1$ is not satisfied. Consequently, we aim to generalize Scott's method, making it still applicable if $\gcd(h_1, h_2) \neq 1$.

Let E denote a pairing-friendly curve admitting a twist E' . Using the above notations, we introduce the generalization of Scott's method, which is summarized as the following theorem.

Theorem 2. Let $\gcd(h_1, h_2) = s$. Define $\{P_1, P_2\}$ to be a basis such that

$$\langle P_1, P_2 \rangle = E'[s].$$

1. If $s|h_2$, assume that $P_1 \in E'(\mathbb{F}_{p^e})[s]$. If the point $Q \in E'(\mathbb{F}_{p^e})$ satisfies that $t_s(P_1, Q) \neq 1$, then $Q \notin \mathbb{G}_2$. Otherwise, it can be deduced that

$$Q \in \mathbb{G}_2 \Leftrightarrow Q \in E'(\mathbb{F}_{p^e}) \text{ and } \psi(Q) = [t - 1]Q.$$

2. If $s^2|h_2$, and the point $Q \in E'(\mathbb{F}_{p^e})$ satisfies that $t_s(P_1, Q) \neq 1$ or $t_s(P_2, Q) \neq 1$, then we have $Q \notin \mathbb{G}_2$. Otherwise, it can be deduced that

$$Q \in \mathbb{G}_2 \Leftrightarrow Q \in E'(\mathbb{F}_{p^e}) \text{ and } \psi(Q) = [t - 1]Q.$$

Proof. If $s = 1$, then it exactly corresponds to Scott's method. Now we consider the case $s > 1$. There are the following two cases:

1. If $s|h_2$, then $E'(\mathbb{F}_{p^e})[s] \cong \mathbb{Z}_s$. Thus there exists a point $P_1 \in E'(\mathbb{F}_{p^e})$ such that $E'(\mathbb{F}_{p^e})[s] = \langle P_1 \rangle$. In other words, arbitrary point $P \in E'(\mathbb{F}_{p^e})[s]$ can be represented as $P = [c]P_1$ where c is an integer. If the reduced Tate pairing $t_s(P_1, Q) \neq 1$, by Lemma 1 we obtain $Q \notin sE'(\mathbb{F}_{p^e})$, which yields that $s_1|\text{ord}(Q)$ where s_1 is a factor of s . Hence, we deduce that $Q \notin \mathbb{G}_2$. If $t_s(P_1, Q) = 1$, then for any point $P \in E'(\mathbb{F}_{p^e})[s]$ we have

$$t_s(P, Q) = t_s([c]P_1, Q) = t_s(P_1, Q)^c = 1.$$

By Lemma 1 we can deduce that $Q \in sE'(\mathbb{F}_{p^e})$. Therefore, the cofactor h_2 can be reduced to $h'_2 = \#sE'(\mathbb{F}_{p^e})/r = h_2/s$. From $\gcd(h_1, h_2) = s$ it can be deduced that $\gcd(h_1, h'_2) = \gcd(h_1, h_2)/s = 1$. Consequently, we can now use Scott's method to execute the testing.

2. If $s^2|h_2$, we obtain $E'[s] \subseteq E'(\mathbb{F}_{p^e})$. Hence, there exists a basis P_1, P_2 such that $E'(\mathbb{F}_{p^e}) = \langle P_1, P_2 \rangle$. Thus for arbitrary point $P \in E'(\mathbb{F}_{p^e})[s]$, it can be represented as $P = [c_1]P_1 + [c_2]P_2$ where $c_1, c_2 \in \mathbb{Z}$. If $t_s(P_1, Q) \neq 1$ or $t_s(P_2, Q) \neq 1$, by Lemma 1 we have $Q \notin sE'(\mathbb{F}_{p^e})$, which yields that $Q \notin \mathbb{G}_2$. If $t_s(P_1, Q) = 1$ and $t_s(P_2, Q) = 1$, then for any $P \in E'(\mathbb{F}_{p^e})[s]$ we obtain

$$t_s(P, Q) = t_s([c_1]P_1 + [c_2]P_2, Q) = t_s(P_1, Q)^{c_1} t_s(P_2, Q)^{c_2} = 1.$$

Consequently, we deduce that $Q \in sE'(\mathbb{F}_{p^e})$. The remaining part of the proof is similar to the case $s||h_2$.

On the basis of Theorem 2, we propose Algorithm 2 for $\mathbb{G}_2$ subgroup membership testing. Here we only consider the situation $s||h_2$ when $s > 1$, and the other is similar.

Note that by performing the above adjustment, our method can be employed even though $\gcd(h_1, h_2) \neq 1$. We also point out that DLZZ criterion is also not limited by the condition $\gcd(h_1, h_2) = 1$. In the following, we show the difference between DLZZ criterion and Theorem 2. Our method accomplishes the membership testing by determining whether the point is in $[s]E'(\mathbb{F}_{p^e})$, while DLZZ criterion makes use of LLL's algorithm to find a suitable short vector and requires to satisfy an additional weaker condition. In certain cases, Algorithm 2 will be more efficient than DLZZ criterion. It is worth noting that the generator(s) P_i of $E'(\mathbb{F}_{p^e})[s]$ can be precomputed. According to Theorem 2, we can see that the corresponding operations of our approach may include the computations of pairing(s) $t_s(P_i, Q)$ and a scalar multiplication by $t - 1$. If s is small, the pairing computation can be considered as the determination of the power- s residue symbol, thus the total cost is close to the method in [39].

We provide the following example (BLS27 family) for description, and verify that the above new method will be more efficient than the algorithm in [16] in some specific cases.

Example 3 (BLS27 family). The BLS27 family with embedding degree $k = 27$ and CM discriminant $D = 3$ is parametrized by the following three polynomials

$$\begin{aligned} r(x) &= \frac{1}{3}(x^{18} + x^9 + 1), \\ t(x) &= x + 1, \\ p(x) &= \frac{1}{3}(x - 1)^2(x^{18} + x^9 + 1) + x. \end{aligned}$$

A BLS27 curve admits a degree-3 twist E' defined over $\mathbb{F}_{p^9}$. We first consider DLZZ criterion [16]. By calculation, the $\gcd$ of the resultant and h_2 in Theorem 1 in [16] is 3, which means that the condition is not satisfied. In [3], the authors

Algorithm 2 Generalization of Scott's membership testing method

Input: The point $Q \in E'(\mathbb{F}_{p^e})$ for the testing. The cofactors h_1 and h_2 . A precomputed s -torsion point P , where $s = \gcd(h_1, h_2)$.

Output: Whether Q is in $\mathbb{G}_2$.

```

1: if  $s = 1$  then
2:   if  $\psi(Q) = [t - 1]Q$  then
3:     return true
4:   else
5:     return false
6:   end if
7: else
8:    $T \leftarrow t_s(P, Q)$ 
9:   if  $T \neq 1$  then
10:    return false
11:  else
12:    if  $\psi(Q) = [t - 1]Q$  then
13:      return true
14:    else
15:      return false
16:    end if
17:  end if
18: end if

```

make a modification and deduce that the corresponding $\mathbb{G}_2$ membership testing is to determine whether

$$[(1 - x)/3](S - \psi^9(S)) + \psi^9(Q) = \mathcal{O}_{E'}$$

for the point $Q \in E'(\mathbb{F}_{p^9})$, where $S = R + \psi^3(R) + \psi^6(R)$, $R = Q + \psi(Q) + \psi^2(Q)$. Consequently, it requires a scalar multiplication $[(1 - x)/3]$, six point additions and six actions of the powers of ψ .

Now we consider the computational cost of our method. By calculation, we obtain

$$\gcd(h_1, h_2) = 3 \text{ and } 3 \parallel h_2.$$

By Theorem 2, the 3-torsion subgroup $E'(\mathbb{F}_{p^9})[3]$ satisfies

$$E'(\mathbb{F}_{p^9})[3] \cong \mathbb{Z}_3.$$

Therefore, we can precompute a 3-torsion point $P_3 \in E'(\mathbb{F}_{p^9})$ such that $\langle P_3 \rangle = E'(\mathbb{F}_{p^9})[3]$, and determine whether

$$t_3(P_3, Q) = (\ell_{P_3, P_3}(Q))^{\frac{p^9-1}{3}} = \left(\frac{\ell_{P_3, P_3}(Q)}{p^9} \right)_3 = 1.$$

Since the slope $\lambda_\ell = \frac{3x_{P_3}^2}{2y_{P_3}}$ of the tangent line ℓ_{P_3, P_3} can also be precomputed, the computation of the above residue symbol can be accomplished by applying cubic

reciprocity law and performing a multiplication over $\mathbb{F}_{p^9}$, which is extremely fast. If $\left(\frac{\ell_{P_3, P_3}(Q)}{p^9}\right)_3 = 1$, Scott's method can be utilized to finish the testing. Otherwise, it yields that $Q \notin \mathbb{G}_2$. Hence, our method needs to perform at most a scalar multiplication $[x - 1]$, an action of ψ and a cubic residue symbol computation.

Table 2 shows the corresponding comparison of the operations between DLZZ criterion [16] and our algorithm on the BLS27 family. Note that our method reduces about 6 point additions in $E'(\mathbb{F}_{p^9})$ compared to DLZZ criterion, as the lengths of the scalar multiplications are almost the same and the other executions are extremely fast. Consequently, our algorithm is more efficient than DLZZ criterion on BLS27 curves, especially for curve BLS27-426 associated with the seed $x = -0\mathbf{x}287\mathbf{beb}$, which possesses the smallest characteristic p among the pairing-friendly curves at the 192-bit security level [3]. In the following, we make a detailed cost comparison for $\mathbb{G}_2$ membership testing on BLS27-426.

By applying NAF, the scalar multiplication $[(1 - x)/3]$ requires to execute 21 point doublings and 7 point additions, while $[x - 1]$ needs to perform 22 point doublings and 7 point additions. Since the action of ψ and the 3-rd residue symbol computation are significantly efficient, the total costs of the testing approach in [16] and our generalized method are about 21 point doublings and 13 point additions, and 22 point doublings and 7 point additions, respectively. From [9], we know that the costs for executing a point addition and a point doubling on $E'(\mathbb{F}_{p^9})$ in Jacobian coordinates are 7 multiplications with 4 squarings, and 2 multiplications with 5 squarings over $\mathbb{F}_{p^9}$, respectively. Hence, it requires to execute 133 multiplications with 157 squarings, and 93 multiplications with 138 squarings over $\mathbb{F}_{p^9}$ by DLZZ criterion and our approach, respectively. This implies that our generalization requires less computation compared to DLZZ criterion.

Table 2. The operations involved in our method and the approach in [16] for the $\mathbb{G}_2$ membership testing on the curve BLS27-426. The notation PADD denotes the point additions.

Family	This work	[16]
BLS27	1 scalar $[x - 1]$	1 scalar $[\frac{1-x}{3}]$
	1 action of ψ	6 actions of powers of ψ
	1 cubical residue symbol $\left(\frac{\ell_{P_3, P_3}(Q)}{p^9}\right)_3$	6 PADD

Although the above generalization is suitable for any pairing-friendly curves in theory, it is cumbersome in some situations. If $\gcd(h_1, h_2) = s$ is large, the computation of the reduced Tate pairing is crucially expensive, making it less efficient than the previous method. In [35], the author recalled that today the s -th power residue symbol (the reduced Tate pairing $t_s(P, Q)$) can be computed in the (sub-)quadratic time only for $s \leq 11$. Besides, on some curves that are not constructed by the BLS method, the size of the scalar $[t(x) - 1]$ is relatively large.

Therefore, in the following, we investigate obtaining a method that requires a shorter scalar multiplication to perform for more generic pairing-friendly curves besides KSS/BLS curves.

3.3 Generalizing DLZZ criterion by utilizing arbitrary degree endomorphisms

In this subsection, we extend DLZZ criterion to all GLV-curves admitting twists by utilizing the degree- n endomorphisms, which is summarized in the following theorem.

Theorem 3. *With the notations as above, let E denote an ordinary curve over $\mathbb{F}_p$ with embedding degree k and CM-discriminant D . Define E' to be a degree- d twist of E over $\mathbb{F}_{p^e}$ such that $r \parallel E'(\mathbb{F}_{p^e})$. Let $\Phi = \phi^{-1} \circ \sigma \circ \phi \in \text{End}(E')$, where $\sigma = a + b\sqrt{-D}$, $a, b \in \mathbb{Q}$ is a degree- n endomorphism on E . Denote by $\text{Res}(h, g)$ the resultant of two polynomials h and g .*

1. *If $\sqrt{-D} \in \mathbb{Q}(\zeta_k)$, it suffices to employ $\psi = \phi^{-1} \circ \pi \circ \phi$ to construct a dimension- $\varphi(k)$ modular lattice*

$$\mathcal{L}_\psi = \{(c_0, \dots, c_{\varphi(k)-1}) \in \mathbb{Z}^{\varphi(k)} \mid \sum_{i=0}^{\varphi(k)-1} c_i \cdot p^i \equiv 0 \pmod{r}\}.$$

Assume that $(c_0, c_1, \dots, c_{\varphi(k)-1}) \in \mathcal{L}_\psi$ satisfies $\gcd(\text{Res}(h(\psi), g(\psi)), h_2 \cdot r) = r$, where $h(\psi) = \sum_{i=0}^{\varphi(k)-1} c_i \psi^i$, $g(\psi) = \psi^2 - t\psi + p$. Given $Q \in E'(\mathbb{F}_{p^e})$, then $Q \in \mathbb{G}_2 = E'(\mathbb{F}_{p^e})[r]$ if and only if

$$\sum_{i=0}^{\varphi(k)-1} [c_i] \psi^i(Q) = \mathcal{O}_{E'}.$$

2. *If $\sqrt{-D} \notin \mathbb{Q}(\zeta_k)$, along with $a \neq 0$ or $k \not\equiv 0 \pmod{4}$, we employ the composed map $\tau = \Phi \circ \psi = \phi^{-1} \circ \sigma \circ \pi \circ \phi$ to construct a dimension- $2\varphi(k)$ modular lattice.*

$$\mathcal{L}_\psi = \{(c_0, \dots, c_{2\varphi(k)-1}) \in \mathbb{Z}^{2\varphi(k)} \mid \sum_{i=0}^{2\varphi(k)-1} c_i \cdot \lambda^i \equiv 0 \pmod{r}\},$$

where λ is a integer such that $\tau(Q') = [\lambda]Q'$ for any $Q' \in \mathbb{G}_2$. Assume that the vector $(c_0, c_1, \dots, c_{2\varphi(k)-1})$ satisfies $\gcd(\text{Res}(h(\tau), g(\tau)), h_2 \cdot r) = r$, where $h(\tau) = \sum_{i=0}^{2\varphi(k)-1} c_i \tau^i$, $g(\tau) = \tau^2 + (at + bfD)\tau + np$. Given $Q \in E'(\mathbb{F}_{p^e})$, then $Q \in \mathbb{G}_2 = E'(\mathbb{F}_{p^e})[r]$ if and only if

$$\sum_{i=0}^{2\varphi(k)-1} [c_i] \tau^i(Q) = \mathcal{O}_{E'}.$$

Proof. If $\sqrt{-D} \in \mathbb{Q}(\zeta_k)$, the effect of $\Phi \sim a + b\sqrt{-D}$ in the composite map $\Phi \circ \psi \sim (a + b\sqrt{-D}) \cdot \zeta_k$ will be killed by $\psi = \phi^{-1} \circ \pi \circ \phi \sim \zeta_k$. Therefore, we only need to utilize the map ψ to construct the modular lattice. The remaining proof is the same as [16, Theorem 1].

We now present the proof for the second situation. If $\sqrt{-D} \notin \mathbb{Q}(\zeta_k)$, define K to be the number field $\mathbb{Q}(\zeta_k \cdot (a + b\sqrt{-D}))$. It follows from [45, Theorem 2.4] that $K = \mathbb{Q}(\zeta_k, \sqrt{-D})$ if $a \neq 0$ or $k \not\equiv 0 \pmod{4}$. Thus the minimal polynomial of $\zeta_k \cdot (a + b\sqrt{-D})$ satisfies that

$$\deg(f(x)) = [\mathbb{Q}(\zeta_k, \sqrt{-D}) : \mathbb{Q}] = 2\varphi(k).$$

By $\tau(Q') = [\lambda]Q'$ for $Q' \in \mathbb{G}_2$ we deduce that $f(\lambda)(Q') = \mathcal{O}_{E'}$, which implies that $f(\lambda) \equiv 0 \pmod{r}$. Consequently, there exists a modular lattice

$$\mathcal{L}_\tau = \{(c_0, c_1, \dots, c_{2\varphi(k)-1}) \in \mathbb{Z}^{2\varphi(k)} \mid \sum_{i=0}^{2\varphi(k)-1} c_i \cdot \lambda^i \equiv 0 \pmod{r}\}.$$

By LLL's algorithm [36] we can obtain a short vector $(c_0, \dots, c_{2\varphi(k)-1}) \in \mathcal{L}_\tau$ such that $|c_i| \approx r^{1/2\varphi(k)}$.

If $Q \in \mathbb{G}_2$, it is clear that $\sum_{i=0}^{2\varphi(k)-1} [c_i] \tau^i(Q) = \mathcal{O}_{E'}$. On the contrary, let b_0 and b_1 be two integers such that

$$h(\tau) \equiv b_0 + b_1 \tau \pmod{g(\tau)}.$$

By the properties of resultants, we have

$$\text{Res}(h(\tau), g(\tau)) = \text{Res}(b_0 + b_1 \tau, g(\tau)) = b_0^2 - (at + bfD)b_0b_1 + nb_1^2p.$$

Denote by $\hat{\tau}$ the dual of τ . By Lemma 1, it can be deduced that

$$\begin{aligned} (b_0 + b_1 \hat{\tau})(b_0 + b_1 \tau) &= b_0^2 + b_0 \cdot b_1 \cdot \text{Tr}(\tau) + b_1^2 \cdot \text{Nrd}(\tau) \\ &= b_0^2 - (at + bfD)b_0b_1 + nb_1^2p, \end{aligned}$$

where $\text{Tr}(\tau) = \tau + \hat{\tau}$ and $\text{Nrd}(\tau) = \tau \cdot \hat{\tau}$ are the trace and the norm of τ , respectively. Since $h(\tau)(Q) = \mathcal{O}_{E'}$ and $g(\tau)(Q) = \mathcal{O}_{E'}$, it yields that

$$\begin{aligned} [\text{Res}(h(\tau), g(\tau))](Q) &= [b_0^2 - (at + bfD)b_0b_1 + nb_1^2p]Q \\ &= (b_0 + b_1 \hat{\tau})(b_0 + b_1 \tau)(Q) \\ &= \mathcal{O}_{E'}, \end{aligned}$$

which implies that $\text{ord}(Q) \mid b_0^2 - (at + bfD)b_0b_1 + nb_1^2p$. By the condition

$$\gcd(\text{Res}(h(\tau), g(\tau)), h_2 \cdot r) = r \text{ and } Q \in E'(\mathbb{F}_{p^e}),$$

we derive that $\text{ord}(Q) = r$, which completes the proof.

For finding a short vector on the GLV-curves admitting dimension- $2\varphi(k)$ decompositions as discussed in the second situation in Theorem 3, we can refer to Algorithm 1 in [16].

Remark 1. Note that $\gcd((b_0^2 - (at + bfD)b_0b_1 + nb_1^2p)/r, h_2) = 1$ can be relaxed into $\gcd((b_0^2 - (at + bfD)b_0b_1 + nb_1^2p)/r, h_2) = s$, where s is a small integer if the desired short vector cannot be found. After deriving a short vector satisfying the relaxed condition, we can utilize [24, Theorem 26.3.3] to determine whether s is the factor of $\text{ord}(Q)$, $Q \in E'(\mathbb{F}_{p^e})$. Since s is small, this checking is crucially efficient. Once $Q \in [s]E'(\mathbb{F}_{p^e})$, the techniques in Theorem 3 can be exploited.

Additionally, the idea in Theorem 3 can also be employed to generalize DLZZ criterion for $\mathbb{G}_1$ -membership testing on more pairing-friendly curves.

Based on the above analysis, our generalization enables DLZZ criterion to be compatible with a broader range of GLV-curves. We state the following example to illustrate our extended approach and make a cost comparison.

Example 4 (FM16-488). Consider the following family (FM16) of pairing curves proposed by Fotiadis and Martindale [28] with embedding degree $k = 10$ and CM-discriminant $D = 2$

$$\begin{aligned} r(x) &= x^{16} - x^{12} + x^8 - x^4 + 1, \\ t(x) &= -x^{12} + x^8 - x^4 + 2, \\ p(x) &= \frac{1}{8}(x^{30} - 2x^{26} + 2x^{24} + x^{22} - 2x^{20} + 2x^{16} - 10x^{12} + x^{10} + 10x^8 - 2x^6 \\ &\quad - 8x^4 + x^2 + 8). \end{aligned}$$

By taking the seed $x = 0\mathbf{x}1430\mathbf{c}$, the above three polynomials correspond to the curve FM16-488 with 141-bit security level [28, Table 5]. The previous method is to find a short vector such that $\gcd(\text{Res}(h(\psi), g(\psi)), h_2 \cdot r) = r$. By utilizing [16, Algorithm 1], we obtain the following short vector

$$V = (0, 0, 1, x^4).$$

It implies that the testing requires to perform $\psi^3(Q) + [x^4]\psi^4(Q)$, which needs a scalar multiplication by x^4 , a point addition and four actions of the endomorphism $\psi = \phi^{-1} \circ \pi \circ \phi$. The main computational cost is the scalar multiplication $[x^4]$.

We now apply Theorem 3. By the condition we know that there exists a **degree-2 endomorphism** σ associated with $\sqrt{-2} \in \mathbb{Z}[\sqrt{-2}] = \text{End}_p(E)$. Therefore, the composited map $\tau = \phi^{-1} \circ \sigma \circ \pi \circ \phi$ corresponds to $\alpha = \sqrt{-2} \cdot \zeta_{10} \in \mathbb{Q}(\sqrt{-2}, \zeta_{10})$. Since $\sqrt{-2} \notin \mathbb{Q}(\zeta_{10})$ and $k \not\equiv 0 \pmod{4}$, by Theorem 3 we deduce that τ admits a dimension- $2\varphi(k)$ expansion. By exploiting the technique in [16, Algorithm 1], we obtain a short vector satisfying $\gcd(\text{Res}(h(\tau), g(\tau)), h_2 \cdot r) = r$

$$V = (64u^2 - 15, -8u, -32u^2 + 8, 4u, 16u^2 - 4, -2u, -8u^2 + 2, (32u - 31)u + 8)$$

where $u = \frac{x+4}{8}$. Consequently, we can go through the scalar multiplications in V by performing the computation of the following sequence

$$\begin{aligned} 2u &\rightarrow 4u \rightarrow 8u \rightarrow 8u^2 - 2 \rightarrow 16u^2 - 4 \rightarrow (32u - 31)u + 8 \\ &\rightarrow 32u^2 - 8 \rightarrow 64u^2 - 15, \end{aligned}$$

which requires eight point doublings, five point additions, and two scalar multiplications by u . Thus, the total computational cost becomes eight point doublings, twelve point additions, two scalar multiplications $[u]$ together with seven actions of τ . The most expensive part is two scalar multiplications $[u]$, which can be roughly regarded as the scalar multiplication $[x^2]$. Table 3 illustrates the comparison for the operations required in the approach in [16] and our generalized method on family FM16. Hence, it can be derived that our approach can reduce the bit-length of the scalar multiplication for the $\mathbb{G}_2$ -membership testing from $\log_2(r)/4$ to $\log_2(r)/8$.

Table 3. The operations involved in our method and the approach in [16] for the $\mathbb{G}_2$ membership testing on family FM16. The notations PADD and PDBL denote the point additions and doublings, respectively.

Family	This work	[16]
FM16	2 scalars $[\frac{x+4}{8}]$	1 scalar $[x^4]$
	7 actions of τ	4 actions of ψ
	8 PDBL + 5 PADD	1 PADD

4 Cost analysis and Implementation results

In this section, we first provide a concrete cost analysis for the $\mathbb{G}_2$ membership testing on curve KSS16-330. Furthermore, we present the computational cost calculation and the experimental result of our implementation for the $\mathbb{G}_2$ membership testing on two 128-bit security curves KSS16-329 and KSS16-330, together with curve KSS16-766 at 192-bit security level.

Notations. Let $\mathbf{m}$ represent the multiplication in the base field $\mathbb{F}_p$, with the characteristic p of 6 digits. Denote by $\mathbf{m}_4$, $\mathbf{s}_4$ and $\mathbf{f}_4$ the costs of the multiplication, squaring and Frobenius endomorphism in $\mathbb{F}_{p^4}$, respectively. For simplicity, we omit the calculations for the addition/subtraction and the operations in the subfields of $\mathbb{F}_{p^4}$.

The implementation for the $\mathbb{G}_2$ membership testing on curve KSS16-330 in the library RELIC works with the projective coordinates ($x = X/Z$, $y = Y/Z$). Consequently, we make the cost analysis by employing the projective coordinates. From [9], we deduce that if we work on projective coordinates, it takes $12\mathbf{m}_4$

and $8\mathbf{m}_4 + 3\mathbf{s}_4$ to execute a point addition and a point doubling on the twist $E'(\mathbb{F}_{p^4})$, respectively. Now we consider the action of the endomorphism ψ . Since we are on the projective coordinates, the corresponding cost is $3\mathbf{f}_4$. As for the computation of the quadratic residue symbol, it can be deduced that the cost is about $3\mathbf{f}_4 + 3\mathbf{m}_4$ by exploiting the law of quadratic reciprocity. Additionally, the action of the endomorphism σ is extremely efficient, so we omit it.

According to Section 3.1, the testing algorithm in [16] requires 30 point doublings, 24 point additions and 7 actions of ψ to perform, while our method needs to execute 36 point doublings, 8 point additions, an action of ψ , an action of σ and a quadratic residue symbol computation. Hence, through the above analysis, the total computational cost of DLZZ criterion and our method can be calculated as $528\mathbf{m}_4 + 90\mathbf{s}_4 + 21\mathbf{f}_4$ and $387\mathbf{m}_4 + 108\mathbf{s}_4 + 6\mathbf{f}_4$, respectively. Thus, our algorithm requires fewer operations to perform.

Furthermore, we integrate our method into the cryptographic library RELIC [1] and derive a performance comparison for the computational cost and in terms of clock cycles for $\mathbb{G}_2$ membership testing between utilizing our technique and the previously best approach on three well-known curves KSS16-329, KSS16-330 and KSS16-766. Our code is compiled and benchmarked on Intel(R) Core(TM) i9-12900K 3.20 GHz with TurboBoost and hyperthreading features are disabled. The details of the comparison for the computational cost and clock cycles are illustrated in Table 4. According to the following table, by exploiting our method, the membership testing for $\mathbb{G}_2$ on KSS16-329, KSS16-330 and KSS16-766 are 24.0%, 33.3% and 29.2% faster than using the algorithm in [16] in terms of clock cycles, respectively.

Table 4. The computational cost and benchmarking results of $\mathbb{G}_2$ membership testing on curves KSS16-329 and KSS16-330 at the 128-bit security level, together with KSS16-766 at the 192-bit security level, reported in $\mathbb{F}_p$ -multiplications with the characteristic of 6 digits and $\times 10^3$ CPU clock cycles, respectively. The results of the clock cycles are averaged over 10^4 executions, and the costs $\mathbf{m}_4$, $\mathbf{s}_4$ and $\mathbf{f}_4$ are measured by $9\mathbf{m}$, $6\mathbf{m}$ and $2\mathbf{m}$, respectively [3, Table 9]. The cost of a multiplication over a ground field $\mathbb{F}_p$ with the characteristic of 12 digits (with respect to KSS16-766) is measured by $\frac{2 \cdot 12^2 + 12}{2 \cdot 6^2 + 6} \mathbf{m} \approx 3.846\mathbf{m}$ [3, Sec. 3.4]. Note that the result has been rounded.

Curve	Measurement	This work	DLZZ criterion [16]	Ratio
KSS16-329 [2]	Clock cycles	874	1149	76.0%
	Computational cost	4683 $\mathbf{m}$	5334 $\mathbf{m}$	87.7%
KSS16-330 [28]	Clock cycles	720	1079	66.7%
	Computational cost	4143 $\mathbf{m}$	5334 $\mathbf{m}$	77.7%
KSS16-766 [3]	Clock cycles	4456	6291	70.8%
	Computational cost	30749 $\mathbf{m}$	40729 $\mathbf{m}$	75.5%

5 Conclusion

Subgroup membership testing is relevant for resisting small subgroup attacks. In this work, we investigated the $\mathbb{G}_2$ membership testing on the KSS16 family, and proposed a new efficient method of the testing for $\mathbb{G}_2$ on the curves constructed by the KSS method. Besides, we revisited the previous methods of membership testing for $\mathbb{G}_2$, and generalized these techniques to make them applicable to more generic pairing-friendly curves. We also made an efficient software implementation of $\mathbb{G}_2$ membership testing on three popular curves KSS16-329, KSS16-330 and KSS16-766, and benchmarked our code in the cryptographic library RELIC. The experimental result showed that our algorithm crucially outperforms those in the previous work. In conclusion, our methods may benefit some cryptographic schemes that require $\mathbb{G}_2$ membership testing on KSS16 families.

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